Practice VI – Motor Velocity Control
Part B. I+P Control Strategy

Abstract

Most PID controllers are adjusted on-site; herein, many different types of tuning rules have been proposed in the literature. The proportional term is based on the present error, the integral term depends on past errors, and the derivative term is a prediction of future errors. However, its implementation may not lead to a good system response for real-world systems. The main drawbacks of basic parallel PID controllers are the effects caused by the lack of proportional action on the process. In order to minimize these effects, modification is made to the parallel PID regulator structure. For this reason, in this lab practice, a new control strategy, called I+P control, is proposed to regulate the angular velocity of the QUBE-Servo 3. Hence, analysis of the control signal is performed and compared with the PI controller.

Topics Covered

- Velocity control
- Proportional-integral-derivative control
- Design control according to specifications

Prerequisites

- Hardware interfacing laboratory experiment
- Matlab® script of 2-D line plot

1. Introduction

The transfer function $P(s)$ that relates the angular velocity and the voltage of the system, $\Omega_m(s)/V_m(s)$ obtained in Lab Practice 1 is the following

$$P(s) \quad = \quad \frac{k_t}{J_{eq}R_m s + k_t(k_m + \mu)} \tag{1.1}$$

where k_t represents the torque constant, J_{eq} is the total moment of inertia acting on the motor shaft, R_m corresponds to the terminal resistance, k_m is the back-emf constant of the motor and μ is the coefficient of friction.

Thus, a proportional, integral, and derivative (PID) control can be implemented to regulate the angular velocity of the QUBE-Servo 3. Here, the control law can be mathematically expressed as

$$u(t) \quad = \quad K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{1.2}$$

As it is described in Eq. 1.2, the proportional term is based on the present error, the integral term depends on past errors, and the derivative term is a prediction of future errors. Hence, the feedback control loop is shown in Fig. 1.1.

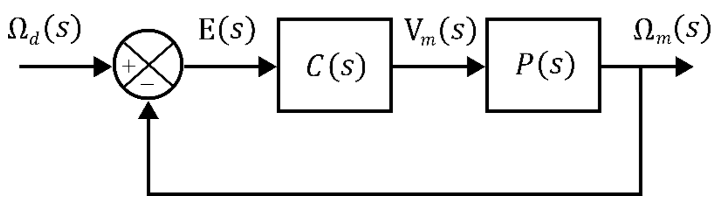


Fig. 1.1: Feedback control loop.

Based on Eq. 1.2 and Fig. 1.1, in Laplace Domain, the controller $C(s) = K_p + \frac{K_i}{s} + K_d s$, it means that the control action is a sum of three terms referred to as proportional (P), integral (I) and derivative (D) control gain. Henceforth, the block diagram of the feedback loop under the PID control strategy is presented in Fig. 1.2.

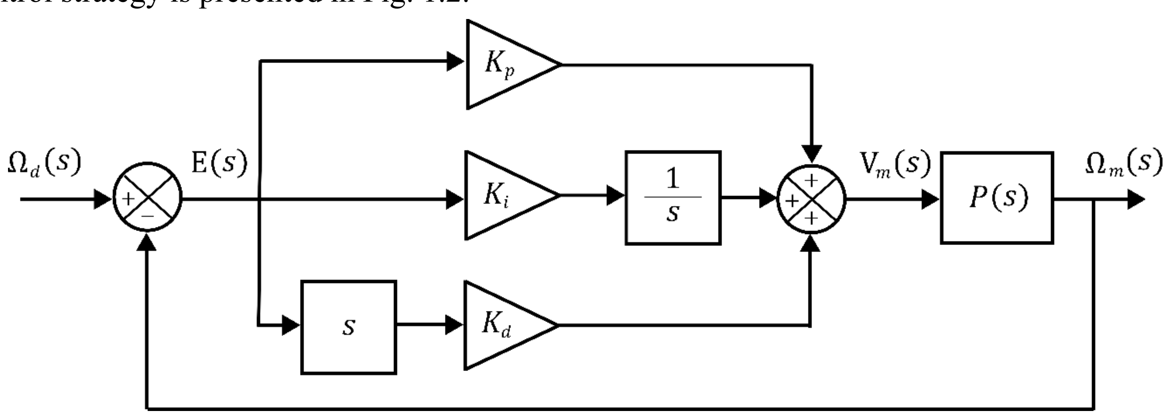


Fig. 1.2: Block diagram of PID control.

Remark 1: The PID controller described by Eq. 1.2 is an ideal PID controller. However, due to measurement noise, this may not lead to a good system response for real world system.

2. Development

D.1 In this Lab Practice, a new control strategy, called I+P control, is proposed to regulate the angular velocity of the QUBE-Servo 3. Here, the derivative term is not used, the integral term takes de error and the proportional considers only the process variable. For this reason, it is also known as integral plus proportional, Fig. 2.1

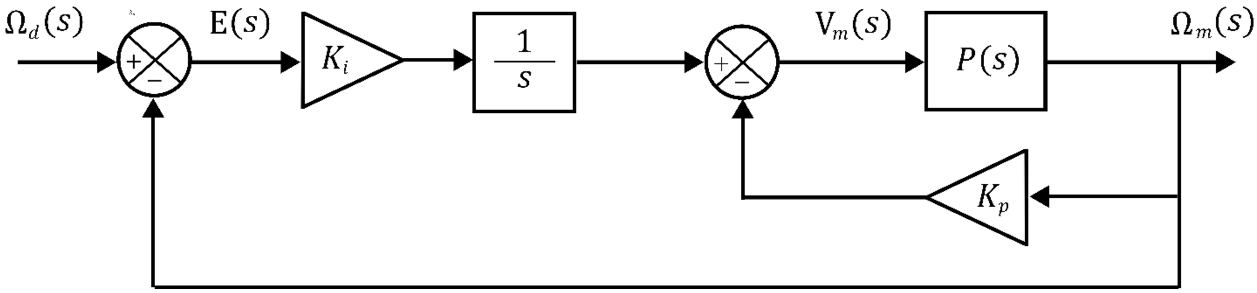


Fig. 2.1: Block diagram of I+P control.

Similarly to Eq. 1.2, obtain the mathematical expression for the control law.

D.2 According to the models already designed in Hardware Interfacing and Filtering laboratory experiment, design a model of the integral plus proportional feedback control (D.1) given in Fig. 2.1 where a Signal Generator block is the velocity command (i.e. desired velocity or setpoint), a square wave with an amplitude of 20 [rad/sec] and at a frequency of 0.25 [Hz], adding a Constant block of 60 [rad/sec], as shown in Fig. 2.2.

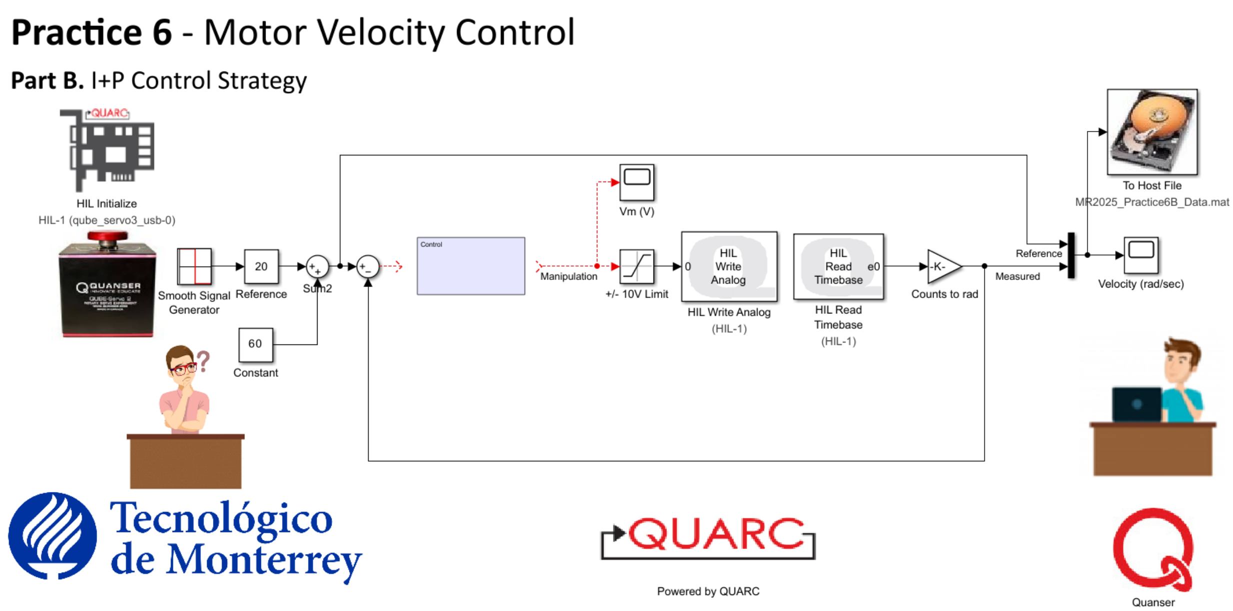
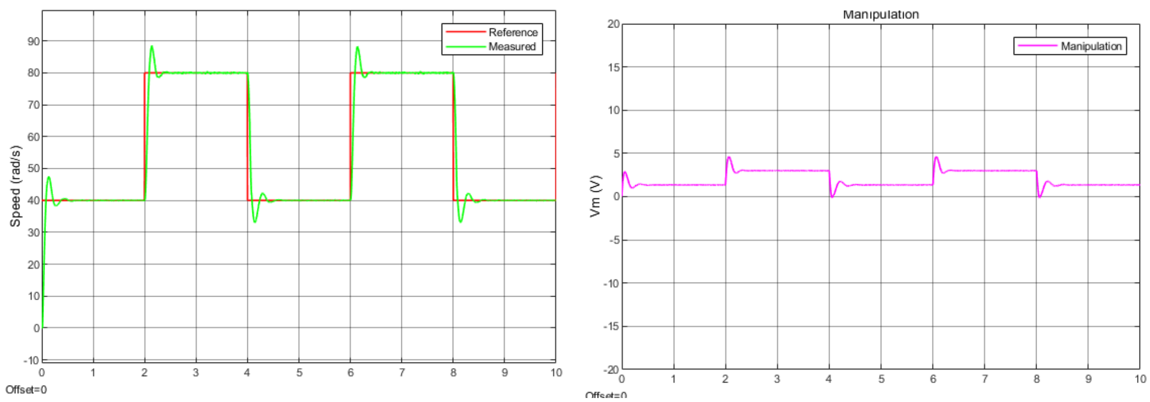


Fig. 2.2: Hardware interfacing and I+P feedback velocity control of QUBE-Servo 3.

D.3 Set the integral gain to $K_i = 2.5[\mathrm{V/(rad-s)}]$ and the proportional gain to $K_p = 0.05[\mathrm{V-s/rad}]$. Build and run the QUARC® controller with your QUBE-Servo 3 model. The scope response should be similar to Fig. 2.3.



a) Motor position b) Applied voltage

Fig. 2.3: Servo control when input voltage to the motor and the QUBE-Servo 3 Model is applied.

D.4 To perform a qualitative analysis of the QUARC® controller given in D.2, run your model of the integral plus proportional feedback control considering the following parameters.

i. Vary $K_i = 1 - 4$ $[\mathrm{V/(rad-s)}]$ and keep $K_p = 0.05$ $[\mathrm{V-s/rad}]$. How does the integral gain affect the velocity control response?

ii. Keep $K_i = 2.5$ $[\mathrm{V/(rad-s)}]$ and vary $K_p = 0 - 0.15$ $[\mathrm{V-s/rad}]$. How does the proportional gain affect the velocity control response?

D.5 Stop the QUARC® controller.

D.6 Based on Fig. 2.1, obtain the closed-loop transfer function $G(s)$ of the system, which relates the actual and the desired angular velocity, $\Omega_m(s)/\Omega_d(s)$.

D.7 Given the QUBE-Servo 3 closed-loop transfer function, find the mathematical expression for the integral K_i and proportional K_p gains in terms of the system's parameters, the natural frequency and the damping factor of the system.

D.8 Based on Table 1.1 of Lab Practice 1, create a Matlab® Script to set the total moment of inertia acting on the motor shaft J_{eq} and the system parameters used in the QUBE-Servo 3 Model.

D.9 According to the QUBE-Servo 3 model parameters, which are the values of the integral K_i and proportional K_p gains of the controller to exhibit an Overshoot of 2.5 % and a Peak Time of 0.15 [sec] when the input is applied?

> **Hint:** Add the code lines to the Matlab® Script developed in D.8 to compute the integral K_i and proportional K_p gains that satisfy the system requirements. Show the mathematical procedure.

D.10 Run the controller with the designed K_i and K_p gains on the QUBE-Servo 3. Attach the velocity response as well as the motor voltage used.

D.11 Based on D.10, measure the Overshoot and Peak Time of the QUBE-Servo 3 response. Do they match the desired percent Overshoot and Peak Time specifications required in D.9 without saturating the motor (going beyond ±10 V)?

D.12 If your response did not match the Overshoot and Peak Time specification, try tuning your control gains until your response does satisfy them. Attach the resulting Matlab® figure, resulting measurements, and comment on how you could modify your controller to arrive at those results.

D.13 By considering the computed gains, regulate the angular position of the QUBE-Servo 3 with a PI control ($K_p = K_p$ and $K_i = K_i$). Hence, the derivative of the error is now considered, Fig. 2.4

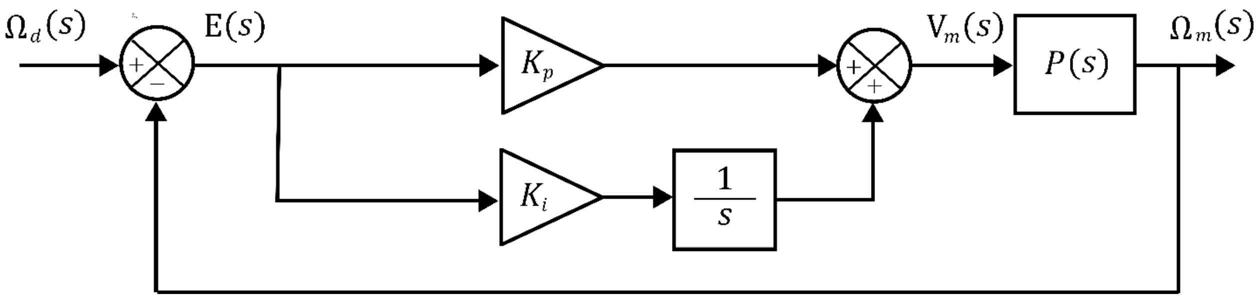


Fig. 2.4: Block diagram of PI control.

What could you observe when a PI controller is implemented?

D.14 Stop the QUARC® controller and power off the QUBE-Servo 3.

3. Report

R.1 Based on D.1, obtain the mathematical expression for the control law, $u(t)$.

R.2 According to D.2 and D.3, show the angle velocity and the input voltage applied to the motor when the integral gain $K_i = 2.5[\mathrm{V}/(\mathrm{rad}-\mathrm{s})]$ and the derivative gain $K_p = 0.05[\mathrm{V}-\mathrm{s}/\mathrm{rad}]$ are set. The designed model of the integral plus proportional feedback control for the QUBE-Servo 3 must be added.

R.3 Based on D.4, show the qualitative analysis of the QUARC® controller developed in D.2.

R.4 Based on D.6, add the mathematical procedure to obtain the QUBE-Servo 3 closed-loop transfer function, $\Omega_m(s)/\Omega_d(s)$. Additionally, according to D.7, show how to get the mathematical expressions of the integral K_i and proportional K_p gains in terms of the system's parameters ω_n and ζ.

R.5 According to D.8 and D.9, show the Matlab® Script generated where the system parameters are defined, and the integral K_i and proportional K_p gains are computed to satisfy the system requirements, $\%MP = 2.5$ % and $t_p = 0.15$ [sec].

R.6 Based on D.10 and D.11, attach the velocity response as well as the motor voltage used under the I+P controller designed. Do they match the desired percent Overshoot and Peak Time specifications required in D.9 without saturating the motor (going beyond ±10 V)?

*Hint:** Support your response graphically.

R.7 Show the tunning procedure of your control gains K_i and K_p performed in D.12 to satisfy the system's requirements.

R.8 What could you observe with the PI controller implemented in D.13?

*Hint:** Add a scope before the saturation block.

R.9 Conclusions.